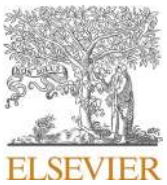
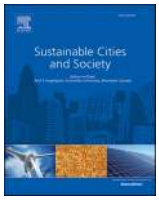
Sustainable Cities and Society 119 (2025) 106083



Contents lists available at ScienceDirect

Sustainable Cities and Society

journal homepage: www.elsevier.com/locate/scs



Exploring temporal and spatial patterns and nonlinear driving mechanism of park perceptions: A multi-source big data study

Xukai Zhao [a,b], He Huang [c], Guangsi Lin [a,b,d,*], Yuxing Lu [e]

[a] State Key Laboratory of Subtropical Building and Urban Science, South China University of Technology, Guangzhou 510641, PR China
[b] Department of Landscape Architecture, School of Architecture, South China University of Technology, Guangzhou, 510641, PR China
[c] Department of Urban Planning, School of Architecture, Tsinghua University, Beijing, 100084, PR China
[d] Guangzhou Key Laboratory of Landscape Architecture, South China University of Technology, Guangzhou, 510641, PR China
[e] Department of Big Data and Biomedical AI, College of Future Technology, Peking University, Beijing, 100091, PR China

ARTICLE INFO

Keywords:
Urban park
Perception
Social media data
Natural language processing
Two-step floating catchment area method
Explainable machine learning

ABSTRACT

To fully realize the benefits of parks, they must be both accessible and usable, with those excelling in these aspects often perceived as more attractive. Traditional surveys for evaluating perceived park accessibility, usability, and attractiveness are expensive and time-consuming, prompting the adoption of social media data as a viable alternative. This study fine-tuned the Chinese-RoBERTa-wwm-ext model on a specially curated dataset to measure perceived accessibility, usability, and attractiveness across 270 parks in Beijing and Guangzhou through 153,872 online comments. We conducted statistical analyses to uncover temporal patterns and incorporate park perception scores into the 2SFCA method for spatial distribution analysis. Additionally, we utilized XGBoost, SHAP, and PDP to investigate the nonlinear driving mechanisms behind these perceptions. Key findings include: (1) Park visitation demonstrates a strong seasonal pattern, with central urban parks consistently outperforming suburban ones; (2) Central subdistricts might face reduced park services due to high population demands; (3) Accessibility is significantly influenced by ticket pricing and transportation availability, especially bus stations; (4) Usability is optimal at a moderate density of sports and fitness facilities (22 per km^2) and proximity to residential areas; (5) Attractiveness benefits from closeness to the Central Business District and amenities such as toilets and restaurants, with a critical park size threshold of 9 km^2. These public-oriented analyses identify areas for improvement and factors shaping public perceptions, providing valuable guidance for strategic decision-making and effective urban management.

1. Introduction

Parks are crucial green spaces in urban environments, providing essential relief from urban sprawl while delivering significant environmental, social, and psychological benefits (Chiesura, 2004). These spaces are pivotal to enhancing urban livability and mitigating the impacts of rapid urbanization, a global trend predicted to house two-thirds of the world's population by 2050 (United Nations, 2018). To fully realize the benefits of parks, they must be both accessible and usable (Wan, Shen et al., 2021; Zhang, Ma et al., 2023), with those excelling in these aspects often perceived as more attractive (Iwarsson & Ståhl, 2003; Wendel, Zarger et al., 2012; Park, 2017; Abdelhamid & Elfakharany, 2020; Li & Lin, 2024; X. Zhao, Lu et al., 2024).

Park accessibility addresses how easily individuals can overcome obstacles such as distance and time to reach and enter a park (Park, 2017; Moran, Rodríguez et al., 2020; Zhang, Ma et al., 2023). It is typically measured by physical access (e.g., proximity and transportation options) and the absence of barriers that hinder visitation (Moran, Rodríguez et al., 2020; Biernacka, Łaszkiewicz et al., 2022; Zhang, Ma et al., 2023; Tian, Wang et al., 2024). Usability reflects how effectively a park meets the functional requirements of various user groups, making it suitable for use (Iwarsson & Ståhl, 2003; Li & Lin, 2024). It encompasses the quality and availability of facilities, alignment of amenities with user needs, and the maintenance of infrastructure (Giles-Corti, Broomhall et al., 2005; Wendel, Zarger et al., 2012; Abdelhamid & Elfakharany, 2020; Moran, Rodríguez et al., 2020). Attractiveness relates to how well a park aligns with individual preferences, influencing both voluntary visits and the duration of stay

* Corresponding author.
E-mail address: asilin@126.com (G. Lin).

https://doi.org/10.1016/j.scs.2024.106083
Received 12 May 2024; Received in revised form 15 December 2024; Accepted 15 December 2024
Available online 16 December 2024
2210-6707/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(Biernacka, Łaszkiewicz et al., 2022; Li & Lin, 2024). Contributing factors include park area, aesthetic qualities, natural features, noise levels, and proximity to commercial facilities (Biernacka, Łaszkiewicz et al., 2022; Zhang, Ma et al., 2023; Zhan, Guo et al., 2024). Assessing these dimensions primarily involves measuring physical attributes, and the findings from such research are crucial for landscape studies aimed at improving park distribution and quality (Abdelhamid & Elfakharany, 2020; Z. Wang, Miao et al., 2021; Biernacka, Łaszkiewicz et al., 2022; Kong, Liu et al., 2022; Zhang, Peng et al., 2022).

As Chiesura (2004) questioned, "How can a city be sustainable if it does not meet the needs of its citizens?" Beyond measurable physical attributes, human perceptions of environments are integral to shaping sustainable urban development. These perceptions represent the psychological experiences and emotional responses individuals have toward their surroundings (Zube, Sell et al., 1982; Xu, Xiong et al., 2023). In this context, perceived park accessibility, usability, and attractiveness are increasingly acknowledged as crucial indicators for understanding park users' preferences, values, attitudes, and behaviors (Wendel, Zarger et al., 2012; Stessens, Canters et al., 2020; Biernacka, Łaszkiewicz et al., 2022; Tang, Zhao et al., 2023). Traditionally, these perceptions have been assessed through self-reported surveys and interviews (Holman, Donovan et al., 1996; Giles-Corti, Broomhall et al., 2005; Wendel, Zarger et al., 2012; Moran, Rodríguez et al., 2020; Stessens, Canters et al., 2020), methods that are often cost-inefficient and time-consuming, particularly when applied to larger study areas.

Recently, social media data (SMD) has emerged as a valuable alternative. User-generated content from platforms such as X (formerly Twitter), TripAdvisor, Google Maps, Dianping, and Ctrip provides a substantial volume of real-time, voluntary feedback, offering insights into users' firsthand experiences, opinions, emotions, and sentiments. By leveraging SMD, researchers can capture public perceptions and human-environment interactions with fewer spatial and temporal restrictions (Cheng, Zhang et al., 2021; Kong, Liu et al., 2022; S. Huai, Liu et al., 2023; Kong, Sarmiento et al., 2023; Rui, 2023; D. Wei, Wang et al., 2024).

Two primary text-mining approaches in Natural Language Processing (NLP) stand out for transforming unstructured online comments into analyzable formats. The first, lexicon-driven analysis, uses predefined lexicons to identify specific elements by calculating keyword frequencies (Z. Wang, Miao et al., 2021; Z. Wang, Zhu et al., 2021; Kong, Sarmiento et al., 2023; Zheng, Pan et al., 2024). However, this approach often neglects context, word meaning, grammatical structure, and word relationships. The second approach, sentiment analysis using advanced deep learning algorithms, converts intangible perceptual information from texts into quantifiable sentiment scores. This allows researchers to evaluate the positive or negative sentiments people expressed about urban environments (Cheng, Zhang et al., 2021; He, Sun et al., 2023; Y. Huai, Liu et al., 2023; D. Wei, Wang et al., 2024). Some studies have further integrated sentiment analysis with word frequency analysis to quantify park satisfaction and explore how environmental features influence public sentiment (Z. Wang, Zhu et al., 2021; Huai & Van de Voorde, 2022; Tang, Zhao et al., 2023; Li, Gao et al., 2024).

Despite accumulated advances, three significant research gaps exist. First, there is a lack of specialized models for evaluating specific perception dimensions using SMD. Due to the complexities involved in constructing dataset and training deep learning models, most studies rely on readily available APIs from NLP platforms for sentiment analysis (Cheng, Zhang et al., 2021; Kong, Liu et al., 2022; S. Huai, Liu et al., 2023; Li, Gao et al., 2024). While these APIs are effective for general sentiment analysis, they fall short in evaluating specific perception dimensions. Consequently, there is a noticeable gap in studies that involve constructing datasets tailored to specific perceptions (e.g., perceived park accessibility, usability, and attractiveness) and fine-tuning advanced deep learning models to accurately assess these dimensions (Zhuang, Qi et al., 2020).

Second, objective obstacles need to be considered in the spatial distribution analysis of park perceptions. Studies using SMD to examine park perceptions typically visualize geographical distribution maps of sentiment scores (Chen, Liu et al., 2018; Cheng, Zhang et al., 2021; Kong, Liu et al., 2022; S. Huai, Liu et al., 2023; Tang, Zhao et al., 2023). While these maps reveal areas with high and low perception scores, incorporating objective obstacles such as travel costs and population density would allow for a more comprehensive evaluation of whether the spatial distribution of parks meets peoples' subjective perceptions while also addressing objective barriers. The Two-step Floating Catchment Area (2SFCA) method, which balances park service supply with population demand while considering travel costs (Luo & Wang, 2003; Xing, Liu et al., 2020), presents a promising solution. However, the integration of park perception scores into this approach remains underexplored.

Third, the exploration of nonlinear relationships between park perceptions and environmental attributes remains limited. The significance of environmental characteristics in shaping park perceptions is well-documented (Byrne & Wolch, 2009; Stessens, Canters et al., 2020). However, existing explorations have predominantly used correlation (Cheng, Zhang et al., 2021; Wan, Shen et al., 2021; Kong, Liu et al., 2022; S. Huai, Liu et al., 2023) and regression analysis (Huai & Van de Voorde, 2022; Li, Gao et al., 2024) to identify linear relationships between environmental attributes and sentiments evaluated form SMD. These linear approaches often fail to capture the complexity of the relationships and may overlook potential effects (Yang, Li et al., 2024). Recent applications of sophisticated machine learning (ML) methods, such as Random Forest (RF) and eXtreme Gradient Boosting (XGBoost), accompanied by explainable ML techniques like Shapley Additive exPlanation (SHAP), have shown great potential to reveal critical environmental factors that have nonlinear or threshold effects on human perceptions (Rui, 2023; Xu, Xiong et al., 2023; Yang, Li et al., 2024).

To address these gaps, this study first fine-tunes advanced deep learning models to effectively measure perceived park accessibility, usability, and attractiveness from SMD. We apply statistical analyses to explore the temporal patterns of these perceptions and incorporate perception scores into the 2SFCA method for spatial distribution analysis. Furthermore, we applied explainable machining learning to examine the nonlinear driving mechanism of park perceptions using multi-source data (Fig. 1 highlights our workflow). These comprehensive analyses of the temporal and spatial patterns of park perceptions, as well as the factors affecting them, provide valuable insights for urban park planning and address three critical research questions:

(1) How can public perceptions of parks be measured using advanced deep learning models?
(2) What are the temporal and spatial patterns of park perceptions?
(3) What are the nonlinear and threshold effects of environmental characteristics on park perceptions?

2. Materials and methods

2.1. Study area

Beijing and Guangzhou (Fig. 2) are highly urbanized and densely populated cities that play critical economic and social roles in northern and southern China, respectively. Both cities face the challenge of an imbalance between population demand and green space supply. According to their respective Statistical Yearbook and Master Urban Plan, available on Beijing's (https://www.beijing.gov.cn/) and Guangzhou's (https://www.gz.gov.cn/) official websites, both cities are committed to the sustainable development of their green spaces. This includes initiatives to promote a balanced distribution and enhance the quality of recreational green spaces. By the end of 2035, Beijing aims to increase its per capita green space from 16.9 m^2 to 17 m^2 and Guangzhou from 17.23 m^2 to 17.5 m^2. Furthermore, both cities are popular tourist destinations, making them excellent subjects for our research given the

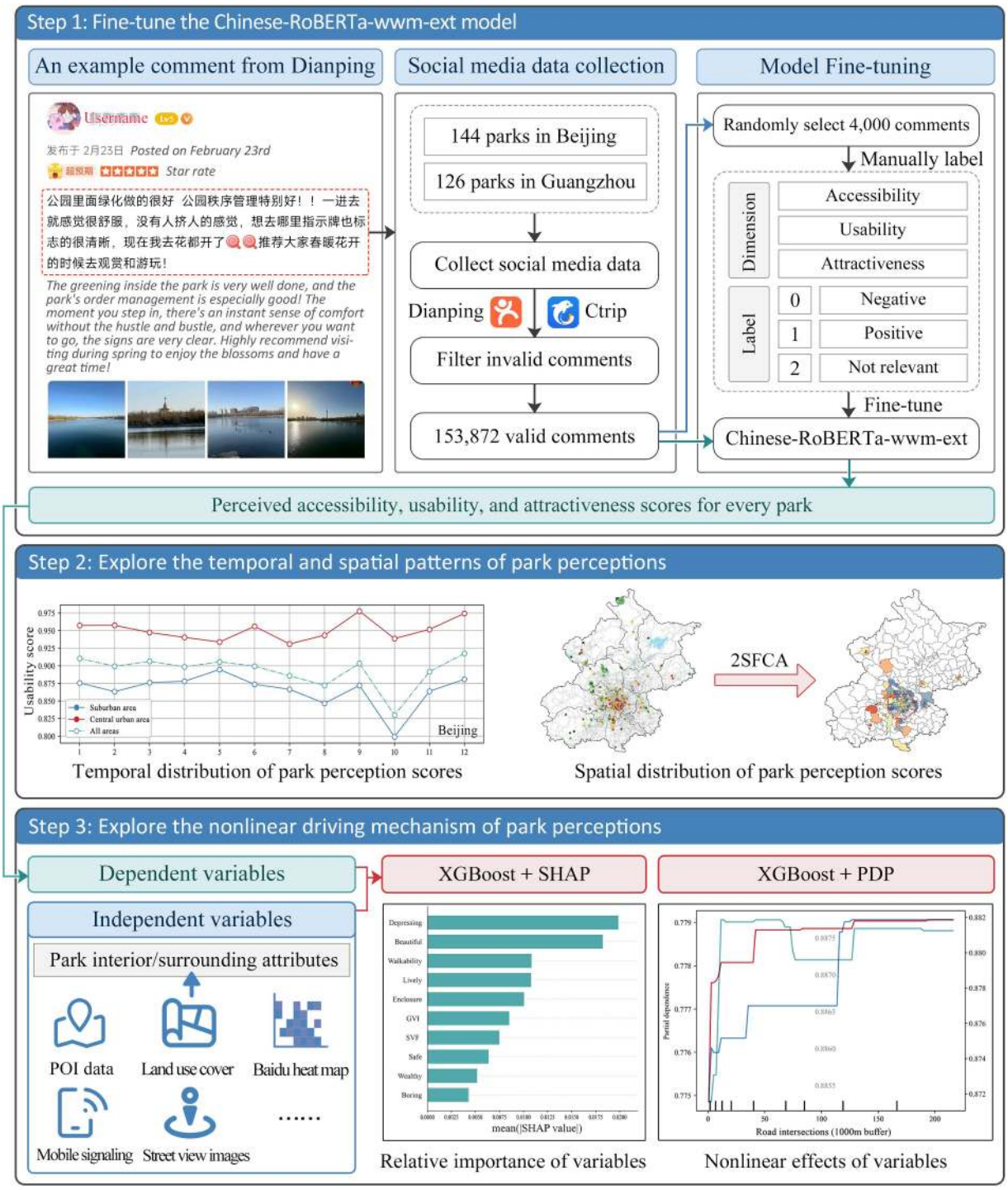


Fig. 1. Research workflow.

abundance of online comments about parks.

2.2. Fine-tuning the Chinese-RoBERTa-wwm-ext model to predict park perceptions

2.2.1. Collection of SMD

We chose Ctrip and Dianping (similar to Yelp and TripAdvisor) as the primary platforms for collecting online comments. In 2023, Ctrip's daily active users reached an impressive 14.22 million, while Dianping's user base peaked at 18.9 million. These platforms allow users to freely express their opinions about parks and interact with others, providing valuable data for park research (Z. Wang, Zhu et al., 2021; Huai & Van de Voorde, 2022; S. Huai, Liu et al., 2023; Li, Fu et al., 2023; Tang, Zhao et al., 2023).

Based on the Park Directories of Beijing (https://yllhj.beijing.gov.cn/ggfw/bjsggml/) and Guangzhou (https://lyylj.gz.gov.cn/), we selected parks that had accumulated >30 comments across both platforms. We utilized the SMD platforms' APIs to collect all comments up until January 1, 2024, culminating in a dataset encompassing 270 parks (144 in Beijing, and 126 in Guangzhou) that met our criteria. After data cleansing to remove blank, meaningless, and duplicate comments, a total of 153,872 valid comments comprising 11,067,318 words were

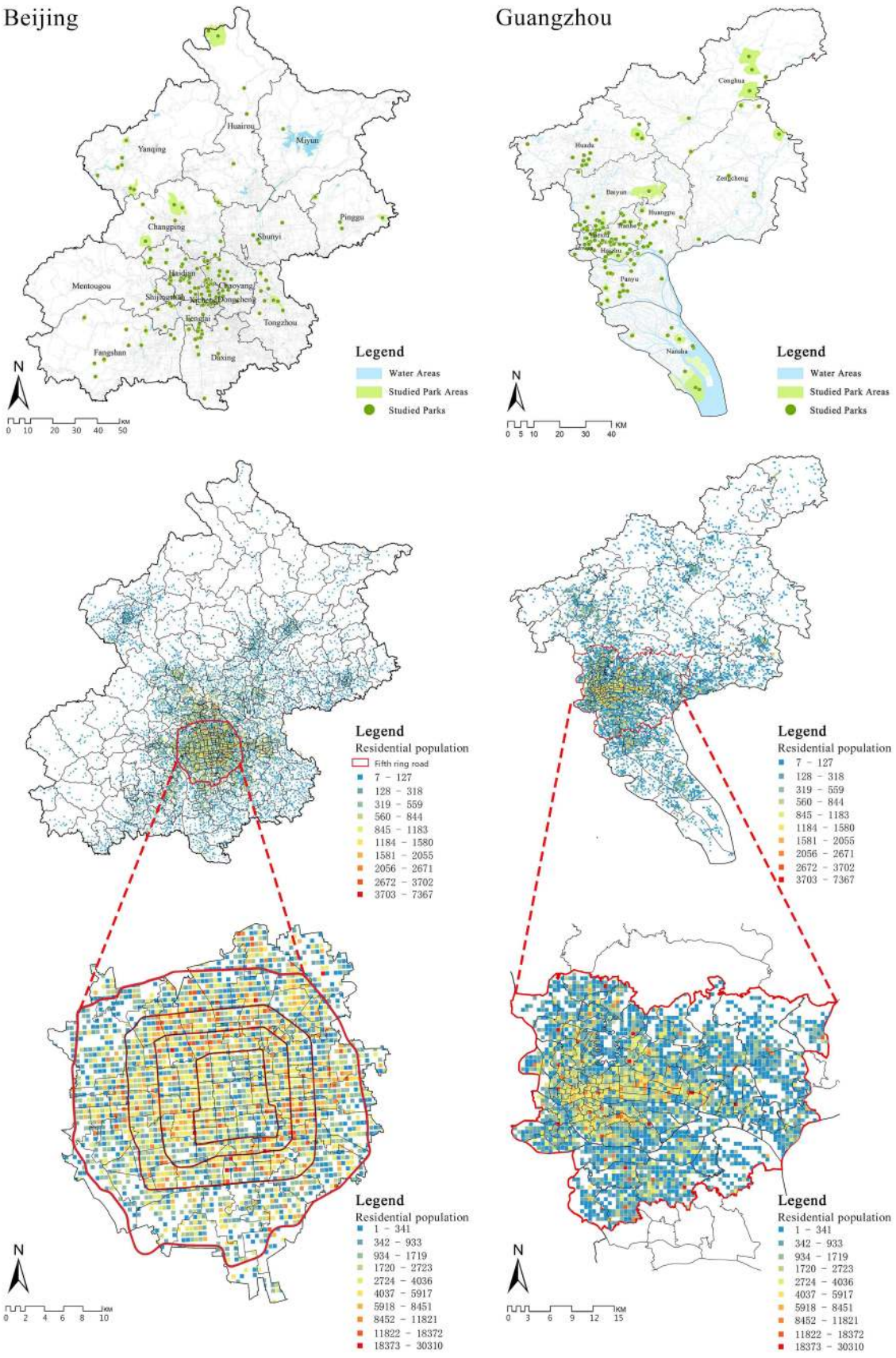


Fig. 2. Spatial distribution of the studied parks and residential population in Beijing and Guangzhou.

obtained. Each comment record includes the username, timestamp, and comment text.

2.2.2. Development of a park perception dataset

We selected a random sample of 4000 comments from those collected in Section 2.2.1. Similar to sentiment analysis, our task involved text classification, but with a more specific focus on the three dimensions of accessibility, usability, and attractiveness. Each comment was manually assessed and assigned labels for these dimensions, with '0′ representing negative feedback, '1′ for positive, and '2′ for irrelevant (the dataset is available in *Appendix A*). This labeling process was meticulously conducted by three experts, and in cases of disagreement, consensus was reached through discussions to ensure consistency and scientific validity.

2.2.3. Fine-tuning the Chinese-RoBERTa-wwm-ext model[f]

The Chinese-RoBERTa-wwm-ext model enhances the original BERT (Bidirectional Encoder Representations from Transformers) architecture by incorporating whole word masking and entity-level masking, along with an extended dataset[g] to better grasp the nuances of the Chinese language, significantly improving the accuracy of Chinese NLP tasks (Cui, Che et al., 2021). We aim to use the fine-tuned model to classify comments into three categories ("0″, "1″, and "2″) for each dimension.

The cross-entropy loss function was utilized to compute the loss, represented by the formula:

$$Loss = -\frac{1}{N}\sum_{i=1}^{N}(y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)) \tag{1}$$

where N is the number of samples, y_i is the ground truth label, and p_i is the predicted probability of the positive class.

We divided the 4,000 manually annotated data into training, validation, and test sets in an 8:1:1 ratio. The data underwent preprocessing to meet the Chinese-RoBERTa-wwm-ext model's input specifications, including tokenization, sequence length normalization through truncation or padding, and label encoding for classification. The model was fine-tuned with a batch size of 32 and a learning rate of 1e-5, using the AdamW optimizer. To avoid under-fitting and over-fitting, we trained the model for 20 epochs, with a linear decay learning rate schedule and a dropout rate of 0.1 within the transformer blocks. For comparative analysis, we selected BERT (He, Sun et al., 2023; Rui, 2023), ERNIE, Long Short Term Memory network (LSTM) (Li, Fu et al., 2023), and XGBoost (Z. Wang, Zhu et al., 2021) as benchmarks, given their prominence in urban sentiment assessment research.

We evaluated the models' performance on the test set using four metrics, including Accuracy (ACC), Precision, Recall, and F1-score. The formulas are as follows:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{TP + FN} \tag{5}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, FN is the number of false negatives.

[f] https://huggingface.co/hfl/chinese-roberta-wwm-ext

[g] The EXT dataset includes Chinese Wikipedia, various encyclopedias, news, Q&A, etc., totaling 5.4 billion words.

2.3. Exploring the temporal and spatial patterns of park perceptions

After fine-tuning the Chinese-RoBERTa-wwm-ext model, we deployed it to predict perception scores for each park across the three dimensions. These scores allowed us to explore the temporal patterns of park perceptions by calculating average monthly scores to illustrate changing trends.

Additionally, we incorporated these scores into the 2SFCA method for spatial distribution analysis. Traditionally, the physical size of a service facility has often been utilized as an indicator of service supply (Rui, 2023; Tian, Wang et al., 2024). This study enhances the 2SFCA method by integrating perception scores into the assessment of the park service capacity A_i^F for each subdistrict.

First, the weighted supply-demand ratio R_j for each park is calculated as follows:

$$R_j = \frac{P_{Accessibility} + P_{Usability} + P_{Attractiveness}}{\sum_{j\in\{d_{kj}\le d_0\}} G(d_{kj},\ d_0) P_k} \tag{6}$$

where d_{kj} represents the distance between the demand point k and the supply place j, which is calculated using actual road networks to estimate travel time. d_0 is the targeted travel time ($d_{kj} \le d_0$). P_k is the residential population of the demand point calculated from mobile signaling data. $G(d_{kj},\ d_0)$ is the Gaussian distance decay function, which quantifies the diminishing effect of distance on service accessibility. The formula for $G(d_{kj},\ d_0)$ is as follows:

$$G(d_{kj},\ d_0) = \begin{cases} \dfrac{e^{-\frac{1}{2}\times\left(\frac{d_{kj}}{d_0}\right)^2} - e^{-\frac{1}{2}}}{1 - e^{-\frac{1}{2}}}, & d_{kj} \le d_0 \\ 0, & d_{kj} > d_0 \end{cases} \tag{7}$$

Subsequently, the park service capacity value A_i^F for each demand point i is calculated as below:

$$A_i^F = \sum_{l\in\{d_{il}\le d_0\}} G(d_{kj},\ d_0) R_l \tag{8}$$

where R_l is the supply-demand ratio of the supply site l within the catchment area ($d_{il} \le d_0$) of the demand location i, and d_{il} quantifies the distance between the demand location i and the center of gravity of the supply location l. A higher value of A_i^F indicates better service capacity.

The assessments considered three travel modes—walking, riding, and driving—within a 20-minute travel time. Ultimately, we employed the Min-Max normalization method to standardize the park service capacity scores.

2.4. Exploring the nonlinear driving mechanism of park perceptions

2.4.1. Variables and measures

This study took the perceived park accessibility, usability, and attractiveness scores as dependent variables and calculated various environmental attributes as independent variables. It incorporated diverse data sources, including official websites, government agencies, and accessible Python APIs, as detailed in *Appendix B*.

Park perceptions are significantly influenced by parks' physical features and conditions, as well as the urban morphology and density of the surrounding environments (Byrne & Wolch, 2009; Chen, Liu et al., 2018; Stessens, Canters et al., 2020; Xing, Liu et al., 2020; D. Wei, Wang et al., 2024). This study focuses on both the internal attributes of parks and the attributes within a 500-meter buffer zone surrounding them (Chen, Liu et al., 2018; Li, Li et al., 2020) (Table 1).

Table 1
Descriptive statistics of all variables.

Categories	Variables	Mean	SD.	Min.	Max.
Dependent variables					
Perception dimensions	Accessibility	0.78	0.17	0.19	1.00
	Usability	0.89	0.11	0.39	1.00
	Attractiveness	0.88	0.10	0.41	1.00
Independent variables					
Park interior attributes	Park area (km^2)	2.36	6.77	0.01	57.77
	Green space ratio	0.60	0.24	0.01	0.99
	Water ratio	0.11	0.15	0.00	0.80
	Building number ratio (quantity/km^2)	1.56	3.10	0.00	27.27
	Building area ratio (m^2/km^2)	0.04	0.06	0.00	0.41
	Road density (m/km^2)	0.19	1.04	0.00	16.40
	Road intersection density (quantity/km^2)	0.01	0.01	0.00	0.09
	Landscape shape index (LSI)	0.71	0.16	0.16	0.93
	Park usability				
	Sports and fitness facility density (quantity/km^2)	8.49	19.96	0.00	198.28
	Toilets density (quantity/km^2)	14.86	23.56	0.00	175.66
	Children's playground density (quantity/km^2)	2.46	7.85	0.00	83.35
	Ticket price (yuan)	10.23	22.11	0.00	150.00
	Distance to CBD (km)	12.87	19.94	0.01	92.72
	Park visitation				
	Average user density (quantity per grid)	9.62	9.11	0.00	74.36
Park surrounding attributes	Building number ratio (quantity/km^2)	5.02	9.75	0.00	69.31
	Building area ratio (m^2/km^2)	0.08	0.15	0.00	1.29
	Vegetation coverage ratio	0.31	0.21	0.00	0.98
	Road density (m/km^2)	9030.35	6384.40	10.21	35274.13
	Road intersection density (quantity/km^2)	403.44	413.69	0.00	2228.00
	Land-use Entropy	0.57	0.37	0.00	1.46
	Accessibility				
	Bus station density (quantity/km^2)	4.29	3.21	0.00	14.20
	Subway station density (quantity/km^2)	6.21	8.59	0.00	53.00
	Usability				
	Hotel accommodation density (quantity/km^2)	11.22	15.75	0.00	86.98
	Restaurant density (quantity/km^2)	29.55	46.62	0.00	650.00
	Parking lot density (quantity/km^2)	181.78	187.75	0.00	1357.00
	Educational facility density (quantity/km^2)	148.22	162.90	0.00	1430.00
	Commercial residence density (quantity/km^2)	113.93	122.29	0.00	726.00
	Economic and demographic characters				
	Working population density (quantity/km^2)	3267.07	3903.68	0.00	28187.73
	Residential population density (quantity/km^2)	2922.79	3101.64	0.00	15134.80
	Average housing price (yuan)	45382.14	37123.18	6640.34	182442.86
	Eye-level street environmental attributes				
	Green View Index (GVI)	0.25	0.07	0.09	0.50
	Sky View Factor (SVF)	0.19	0.09	0.05	0.70
	Enclosure	0.42	0.09	0.10	0.58
	Walkability	0.12	0.05	0.02	0.40
	Beautiful	4.86	0.55	2.00	7.00
	Wealthy	4.60	0.87	1.00	6.56
	Safe	5.62	0.74	1.00	7.00
	Lively	4.43	1.03	1.00	6.23
	Depressing	4.84	0.91	1.00	6.64
	Boring	5.48	0.51	2.00	7.00

(1) Park interior attributes typically include park's design and specific features (Fan, Duan et al., 2021). We analyzed attributes using data from Open Street Map and Baidu Map, which cover park area, green space ratio, water ratio, building number and area ratio, road and road intersection density, and Landscape Shape Index (LSI). LSI is calculated as follows:

$$LSI = \frac{2\sqrt{\pi * A_i}}{C_i} \tag{9}$$

where A_i is the area of $Park_i$, and C_i is the perimeter of the AOI boundary of $Park_i$.

(2) We calculated the density of sports and fitness facilities, toilets, children's playgrounds, bus stations, subway stations, accommodations, restaurants, parking lots, educational facilities, and business residences from Point of Interest (POI) data (Chen, Liu et al., 2018; D. Wei, Wang et al., 2024). This involves dividing the quantity of each POI category by the area of the park or its 500-m buffer zone.

(3) Land-use Entropy was used to measure the diversity of land uses, based on the premise that mixed land uses offer complementary benefits, thus enhancing the utility of each land type (Chen, Liu et al., 2018). The calculation is based on land use cover data, and the formula is as follows:

$$Entropy = -\sum (P_i)(\ln P_i) \tag{10}$$

where P_i is the ratio of the area of the i th land type to the total area. A higher entropy value indicates a greater variety of land uses within the unit.

(4) Following prior studies (Yang, Li et al., 2024), we selected housing price as an economic variable and calculated residential and working population figures from mobile signaling data.

(5) We used street view images (SVIs) from Baidu Map to evaluate the quality of the park surroundings from a pedestrian's viewpoint. Firstly, we employed the Segformer-B5 model (Xie, Wang et al., 2021) trained on the Cityscapes dataset to quantify 19 element categories from SVIs, including roads, buildings,

vegetation, and vehicles, among others. Informed by prior research (Rui, 2023; Yang, Li et al., 2024; X. Zhao, Lu et al., 2024), we calculated Green View Index (GVI), Sky View Factor (SVF), Enclosure, and Walkability using the proportions of various elements (Ma, Ma et al., 2021). The calculation formulas are as follows:

$$GVI_i = \frac{\sum_{j=1}^{4} Vegetation_{ij}}{\sum_{j=1}^{4} N_{ij}} \tag{11}$$

$$SVF_i = \frac{\sum_{j=1}^{4} Sky_{ij}}{\sum_{j=1}^{4} N_{ij}} \tag{12}$$

$$Enclosure_i = \frac{\sum_{j=1}^{4} \left(Building_{ij} + Wall_{ij} + Fence_{ij} + Pole_{ij} + Light_{ij} + Sign_{ij} + Vegetation_{ij}\right)}{\sum_{j=1}^{4} N_{ij}} \tag{13}$$

$$Walkability_i = \frac{\sum_{j=1}^{4} \left(Fence_{ij} + Sidewalk_{ij}\right)}{\sum_{j=1}^{4} Road_{ij}} \tag{14}$$

where N_{ij} represents the total number of pixels in the j-th direction of image i, and $Category_{ij}$ represents the count of pixels belonging to a specific category in the j-th direction of image i.

Further, we applied the "SegFormer-B5 + ConvNeXt-B + Random Forest" approach, trained on the Place Pulse 2.0 dataset (Dubey, Naik et al., 2016) following the methodology from X. Zhao, Lu et al. (2024), to effectively predict six subjective indicators—beautiful, boring, depressing, lively, safe, and wealthy—from SVIs (Fig. 3).

2.4.2. *Identifying nonlinear driving mechanisms through explainable machine learning*

XGBoost stands out in supervised learning for its precision, speed, and simplicity (Chen & Guestrin, 2016). It utilizes gradient boosting frameworks to build predictive models capable of discerning complex, non-linear relationships without predefined assumptions about data

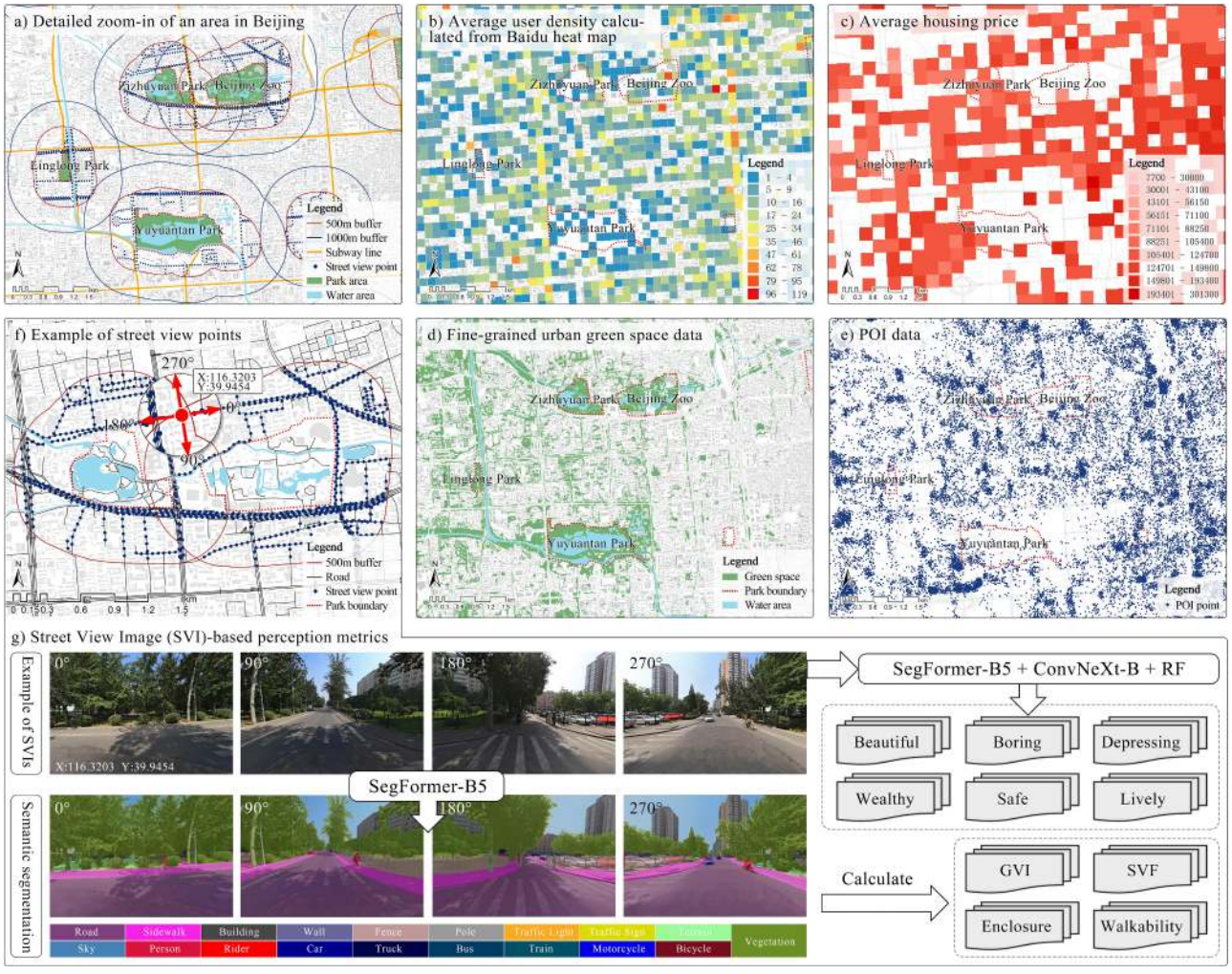


Fig. 3. Data visualizations. Panels (a)-(e) display urban data including user density, housing price, green space, and POI. Panels (f) and (g) illustrate SVIs and the corresponding metrics extracted using deep learning methods.

distribution (Rui, 2023; Wenzhu, Enjia et al., 2024). We divided the dependent and independent variables for 270 parks into training and testing sets with a 7:3 ratio, employing five-fold cross-validation to confirm XGBoost's reliability. Its performance markedly surpasses that of Gradient Boosting Machines (GBM), Random Forest (RF), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN), demonstrated by higher R-square values, and lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) scores (see *Appendix C*).

XGBoost facilitates variable importance ranking through Shapley Additive exPlanations (SHAP) and reveals the non-linear impacts of variables on park perceptions using Partial Dependence Plots (PDP) (Liu, Li et al., 2023, Xu, Xiong et al., 2023) (more details in *Appendix D*).

3. Results

3.1. Training and evaluating the Chinese-RoBERTa-wwm-ext model

Experimental results (Table 2) show that the Chinese-RoBERTa-wwm-ext model outperforms all benchmarks across various metrics, achieving an average accuracy of 85.10 % across three dimensions. It also records high precision and recall, leading to strong F1 scores that indicate well-balanced precision-recall trade-offs. In contrast, the LSTM and XGBoost models display significantly lower performance than the transformer-based models (Chinese-RoBERTa-wwm-ext, BERT, and ERNIE), reflecting the limitations of traditional machine learning approaches for nuanced text classification tasks. The fine-tuned Chinese-RoBERTa-wwm-ext model was further employed to predict the park perception scores across the 153,872 comments of the selected 270 parks.

3.2. Delineating themporal patterns of park perceptions

The count of online comments has been proven to be a reliable proxy for measuring park visitation (S. Huai, Liu et al., 2023). In Beijing, the temporal distribution of these comment counts (Fig. 4) displays a bimodal pattern, with peaks in park visitation during mid-year and a noticeable decline during the colder months, suggesting strong seasonal influences. Conversely, Guangzhou shows a decrease in park visitations from June to September, aligning with the city's hot and humid climate.

Table 2
Comparative performance of the text classification models based on ACC, Precision, Recall, and F1 score ($N = 4000$).

	Model	Accessibility	Usability	Attractiveness
ACC	Chinese-RoBERTa-wwm-ext	0.8338	0.8568	0.8623
	BERT	0.8260	0.8151	0.7870
	ERNIE	0.8156	0.8203	0.8416
	LSTM	0.4714	0.4518	0.5521
	XGBoost	0.4805	0.4818	0.6146
Precision	Chinese-RoBERTa-wwm-ext	0.8380	0.8570	0.8456
	BERT	0.8299	0.8130	0.8011
	ERNIE	0.8083	0.8194	0.8385
	LSTM	0.4502	0.4360	0.4879
	XGBoost	0.4539	0.4577	0.4598
Recall	Chinese-RoBERTa-wwm-ext	0.8338	0.8568	0.8083
	BERT	0.8260	0.8151	0.7870
	ERNIE	0.8086	0.8203	0.8416
	LSTM	0.4714	0.4518	0.5521
	XGBoost	0.4805	0.4818	0.6146
F1 score	Chinese-RoBERTa-wwm-ext	0.8344	0.8558	0.8249
	BERT	0.8270	0.8105	0.7916
	ERNIE	0.8063	0.8198	0.8386
	LSTM	0.4558	0.4410	0.5073
	XGBoost	0.3479	0.3446	0.4754

Note: All models listed here are trained on our park perception dataset.

Both cities exhibit a significant increase in park visitation in October, however, this surge in visitors correlates with a decline in perceptions within suburban areas. Throughout the year, central urban parks in both cities consistently maintain high scores across all dimensions.

3.3. Delineating spatial patterns of park perceptions

3.3.1. Spatial distribution of park perception scores

As shown in Table 3, central urban parks in Beijing exhibited high scores for accessibility (mean = 0.90, SD = 0.07), usability (mean = 0.95, SD = 0.05), and attractiveness (mean = 0.95, SD = 0.04), with the majority (95.43 %) scoring between 0.8 and 1.0. Suburban parks in Beijing, however, show greater variability, particularly in accessibility, with a mean score of 0.76 and a standard deviation of 0.14.

In Guangzhou, while central urban parks exhibit better scores compared to suburban counterparts in all dimensions, they trail behind Beijing's central urban parks. Only 56.79 % of parks in Guangzhou's central areas exceed 0.8 in accessibility, significantly less than the 90.20 % in Beijing's central urban area. Suburban parks in Beijing also outperform those in Guangzhou, with a higher percentage achieving scores between 0.8 and 1.0 in usability (74.19 %) and attractiveness (81.72 %). However, Guangzhou's suburban parks, with 71.11 % scoring above 0.8 in usability and 60 % in attractiveness, still maintain commendable quality levels. Notably, Beijing's suburban areas feature a minority of parks that score below 0.6, accounting for 7.53 % in usability and 3.32 % in attractiveness, highlighting specific areas for targeted enhancement efforts (Fig. 5).

3.3.2. Spatial distribution of park service capacity

The walking maps (Fig. 6a) show that while high-service subdistricts are mainly located in central urban areas, many zones within these areas remain underserved. The riding maps (Fig. 6b) illustrate a noticeable expansion of park service coverage within central areas. Subdistricts outside the city center, such as Rulin, Tantuosi, and Yingfeng in Beijing, and Xinhua, Shatou, Shiqiao, and Wanqingsha in Guangzhou, showcase superior service levels. This highlights the effectiveness of non-motorized transport in enhancing accessibility. The service scope dramatically broadens beyond central urban regions when assessed in driving mode (Fig. 6c). Several peripheral subdistricts in both cities achieve the highest service scores. Notably, service distribution in Beijing shows a significant decline from west to east within the central region, whereas Guangzhou maintains robust service levels throughout its central areas, with the city center demonstrating privileged access to high-quality parks.

3.4. Magnitude and relative importance of effects

3.4.1. Variable importance analysis based on SHAP

Fig. 7 illustrates the importance ranking of various environmental attributes affecting perceived park accessibility, usability, and attractiveness, as quantified by mean SHAP values.

In terms of accessibility, 'ticket' is the most influential interior attribute, followed by 'distance to CBD' and 'average user density'. The leading external factors include the availability of bus stations and parking lots, as well as road density. Usability is primarily influenced by the presence of sports and fitness facilities in parks. Surrounding attributes such as 'commercial residence', 'parking lot', and 'bus station' also play significant roles. Regarding attractiveness, 'distance to CBD', 'park area', the availability of toilets, nearby 'bus station', 'building number ratio', and 'restaurants' are dominant factors.

Interestingly, depressing, lively, and beautiful perceptions of the SVIs all contribute highly to park perceptions. Moreover, 'walkability' is especially influential in enhancing both usability and attractiveness, underscoring the importance of pedestrian-friendly design in park environments.

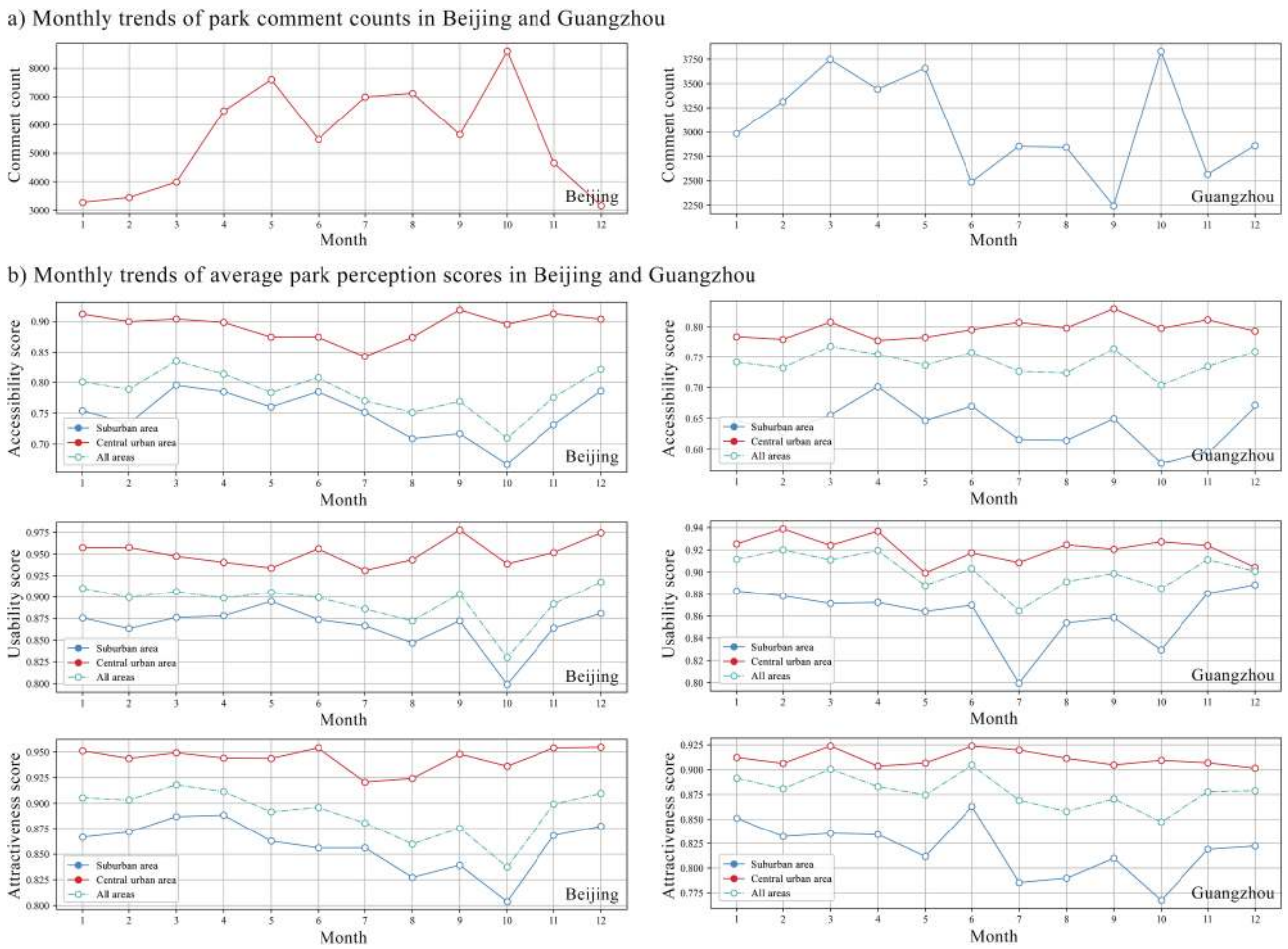


Fig. 4. Monthly trends of comment counts and average park perception scores in Beijing and Guangzhou.

Table 3
Statistical overview of park perception scores.

Area		Metric	Mean	SD.	Min.	Max.	Percentage of Parks (%)				
							0.0–0.2	0.2–0.4	0.4–0.6	0.6–0.8	0.8–1.0
Beijing	Central urban area	Accessibility	0.90	0.07	0.64	1.00	0.00	0.00	0.00	9.80	90.20
		Usability	0.95	0.05	0.77	1.00	0.00	0.00	0.00	3.92	96.08
		Attractiveness	0.95	0.04	0.82	0.99	0.00	0.00	0.00	0.00	100.00
	Suburban area	Accessibility	0.76	0.14	0.39	1.00	0.00	1.08	15.05	36.56	47.31
		Usability	0.85	0.13	0.39	1.00	0.00	1.08	6.45	18.28	74.19
		Attractiveness	0.87	0.11	0.41	0.99	0.00	0.00	3.23	15.05	81.72
Guangzhou	Central urban area	Accessibility	0.81	0.14	0.47	1.00	0.00	0.00	8.64	34.57	56.79
		Usability	0.90	0.09	0.59	1.00	0.00	0.00	1.23	7.41	91.36
		Attractiveness	0.89	0.09	0.54	0.99	0.00	0.00	2.47	13.58	83.95
	Suburban area	Accessibility	0.63	0.22	0.19	1.00	2.22	20.00	22.22	28.89	26.67
		Usability	0.87	0.10	0.68	1.00	0.00	0.00	0.00	28.89	71.11
		Attractiveness	0.82	0.11	0.60	1.00	0.00	0.00	0.00	40.00	60.00

3.4.2. Nonlinear effects of environmental attributes on park perceptions based on PDP

(1) Park interior attributes

The marginal nonlinear effects of environmental attributes on park perceptions were visualized using PDPs generated by the XGBoost model (Figs. 8 and 9). Results reveal that increases in park areas do not uniformly enhance park quality. A distinct threshold at approximately 9 km^2 is observed, beyond which usability significantly declines. Accessibility and attractiveness also show a declining tendency as the park area increases. Conversely, while increases in 'water ratio' or 'green space and water ratio' lead to reduced accessibility, they positively influence the parks' appeal and functionality. The 'green space ratio' exhibits a complex interaction with park perceptions, achieving optimal benefits across all three dimensions at an 80 % ratio.

Furthermore, enhanced provisions of toilets, roads, and buildings within parks are positively correlated with increased usability and attractiveness. However, achieving an optimal balance in the building area and the number of children's playgrounds is crucial, as an oversupply may not yield additional benefits. Similarly, parks equipped with a variety of sports and fitness facilities are perceived as more accessible and user-friendly, but a density exceeding 15 facilities per km^2

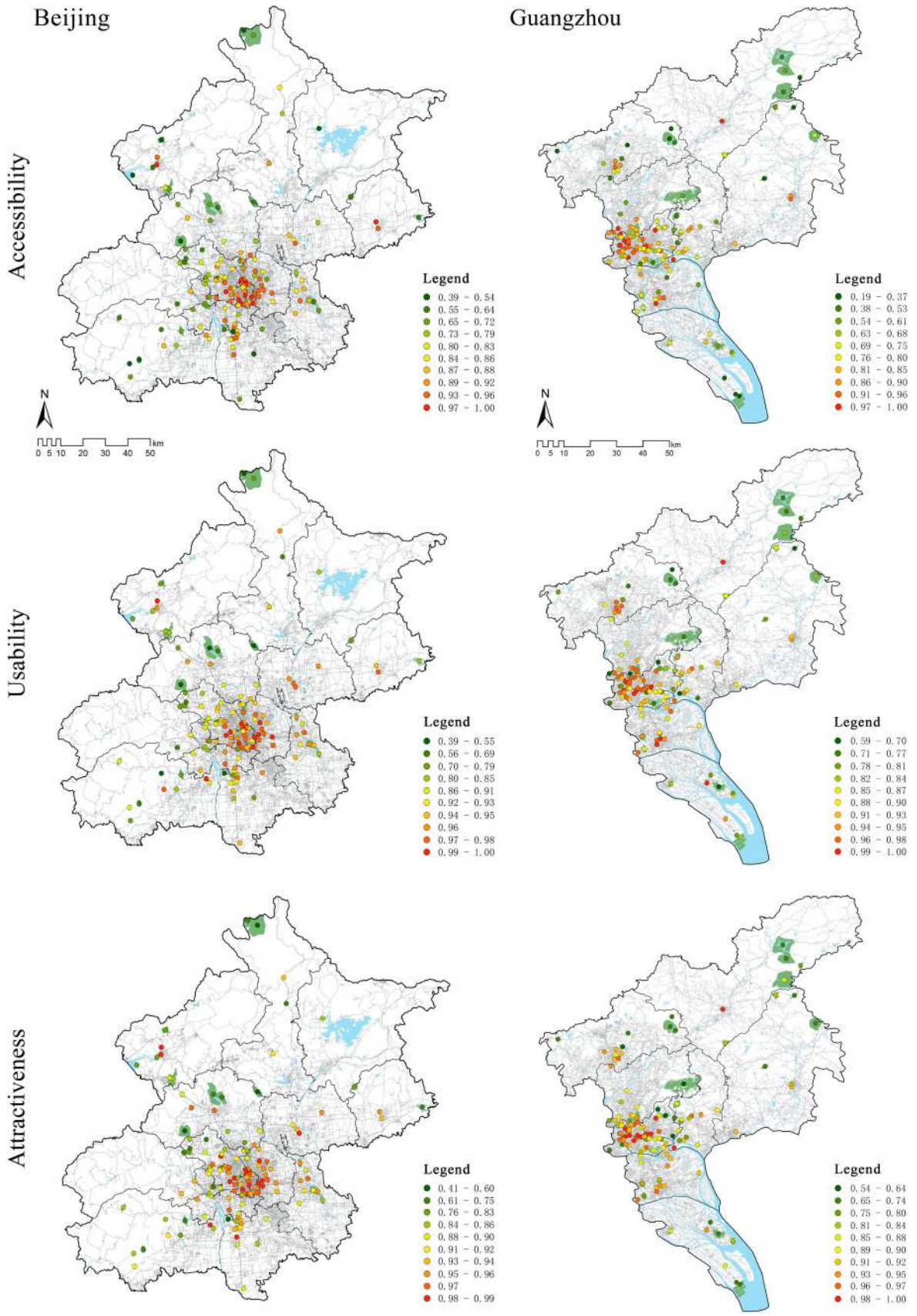


Fig. 5. Geographic distribution maps of park perception scores in Beijing and Guangzhou.

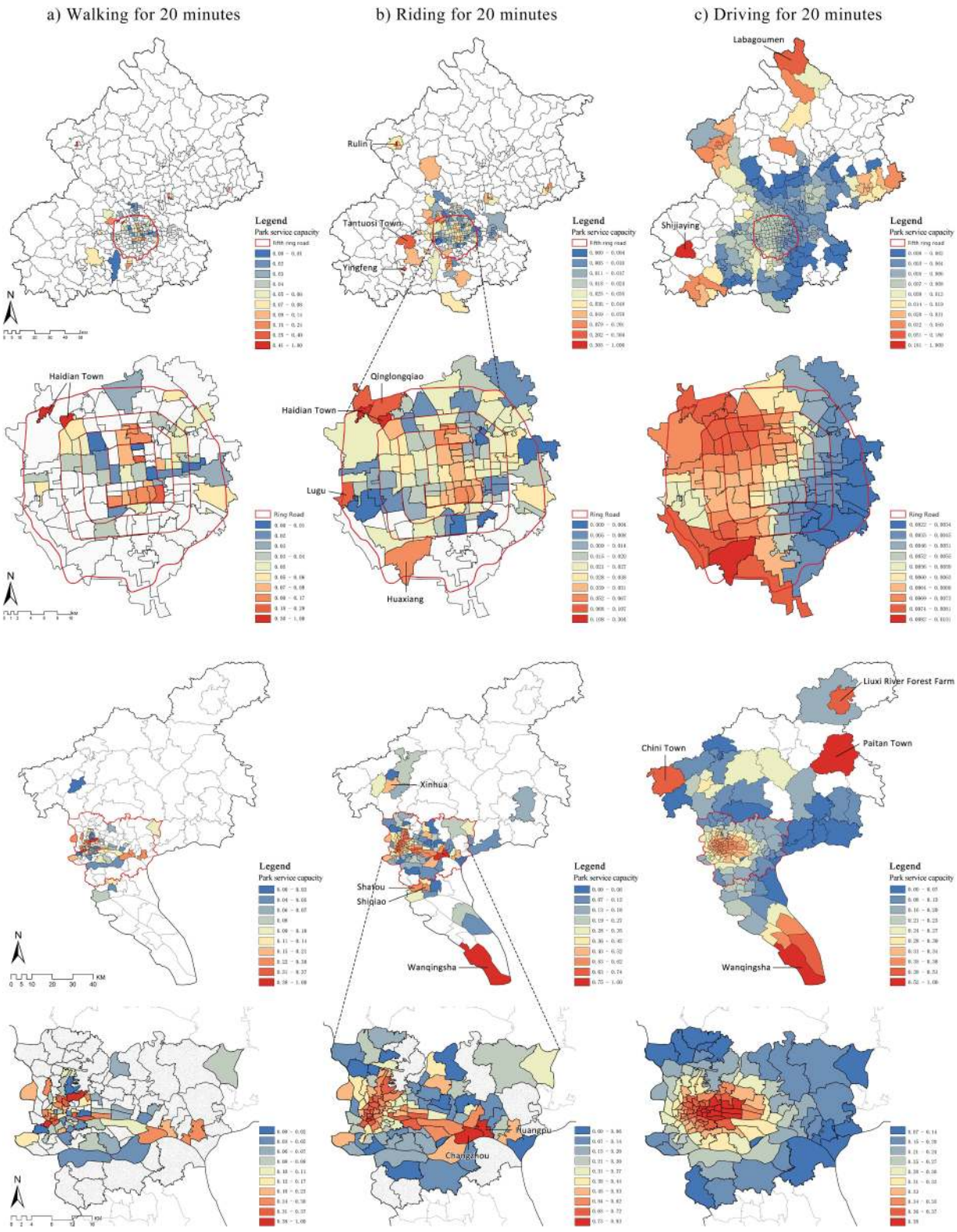


Fig. 6. Distribution maps of park service capacity under three travel modes.

negatively impacts their attractiveness.

Parks with high average user density tend to be more accessible and attractive, but overcrowding can create a mismatch between facility supply and demand, leading to diminished usability (Yang, Li et al., 2024; Zhan, Guo et al., 2024). Parks with simpler perimeters (low LSI value) are easier to access, and those with an LSI between approximately 0.55 and 0.85 offer enhanced usability and attractiveness. Parks offering free entry are often perceived as having high quality. Conversely, parks charging an entry fee exceeding 80 yuan, while maintaining high usability, are generally perceived as less accessible and less appealing. Moreover, parks closer to CBD are associated with higher perception scores.

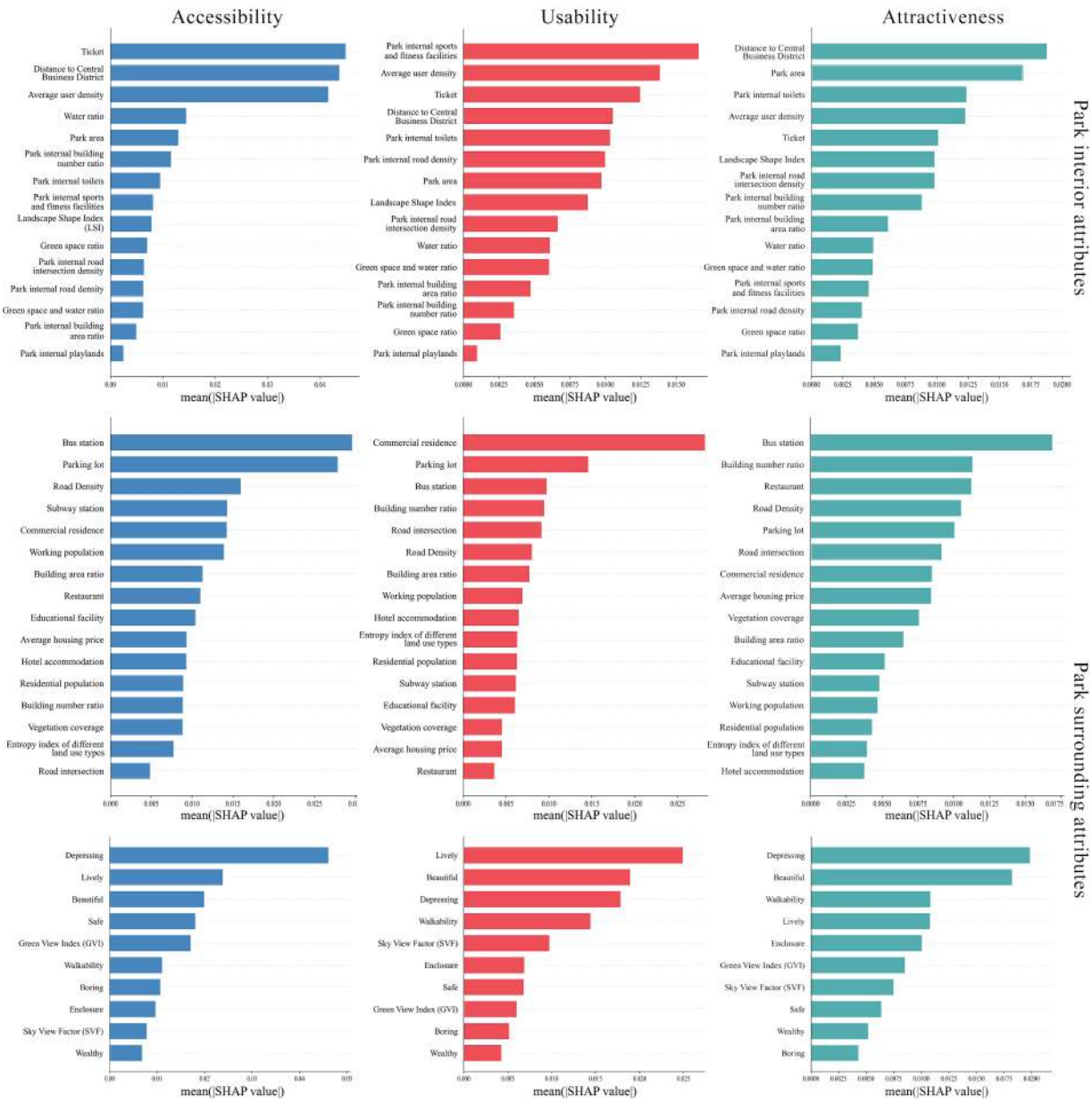


Fig. 7. Results of the relative importance of environmental attributes on park perceptions.

(2) Park surrounding attributes

High vegetation coverage, typically located far from built areas, presents a threshold for accessibility at 40 %. Interestingly, the quantity of buildings within parks positively correlates with usability and attractiveness, whereas a larger building area has the opposite effect (Fig. 9). However, both the number and area of the buildings adjacent to parks enhance perceptions across all three dimensions.

Transportation-related infrastructure, including bus and subway stations, parking lots, road density, and road intersections, is vital for accessibility. Parks in areas with high land-use entropy are more accessible, but this does not significantly boost park use (Chen, Liu et al., 2018), and may reduce park attractiveness.

Facilities such as hotels are shown to enrich park experience. Areas with educational facilities like kindergarten, primary, and secondary schools are associated with greater usability and accessibility. A density threshold of fewer than 10 facilities per km^2 is identified as optimal for maintaining park attractiveness. Besides, the density of commercial residences exhibits a complex impact on park perceptions, positively correlating with usability but following an inverted U-shaped pattern with accessibility and attractiveness, indicating that both low and moderate levels are preferable. Restaurant density is the third important variable deciding attractiveness, with the ideal range between 35 and 50 establishments per km^2.

Regarding population metrics, a working population density of approximately 4000 to 7000 per km^2 around parks maximizes perception quality. Residential populations below 4500 per km^2 improve accessibility and usability, while those below 2000 per km^2 enhance attractiveness. Higher housing prices correlate with improved perceptions of park quality, highlighting an economic disparity (Tian, Wang et al., 2024).

Street view attributes, though not encompassing the internal park environment, show impacts on visitor perceptions. Positive perceptions such as ‘beautiful’, ‘lively’, ‘safe’, and ‘wealthy’ are aligned with favorable park perceptions, whereas negative ones like ‘depressing’ and ‘boring’ are linked to poor perceptions. Walkability positively affects public perception, indicating the

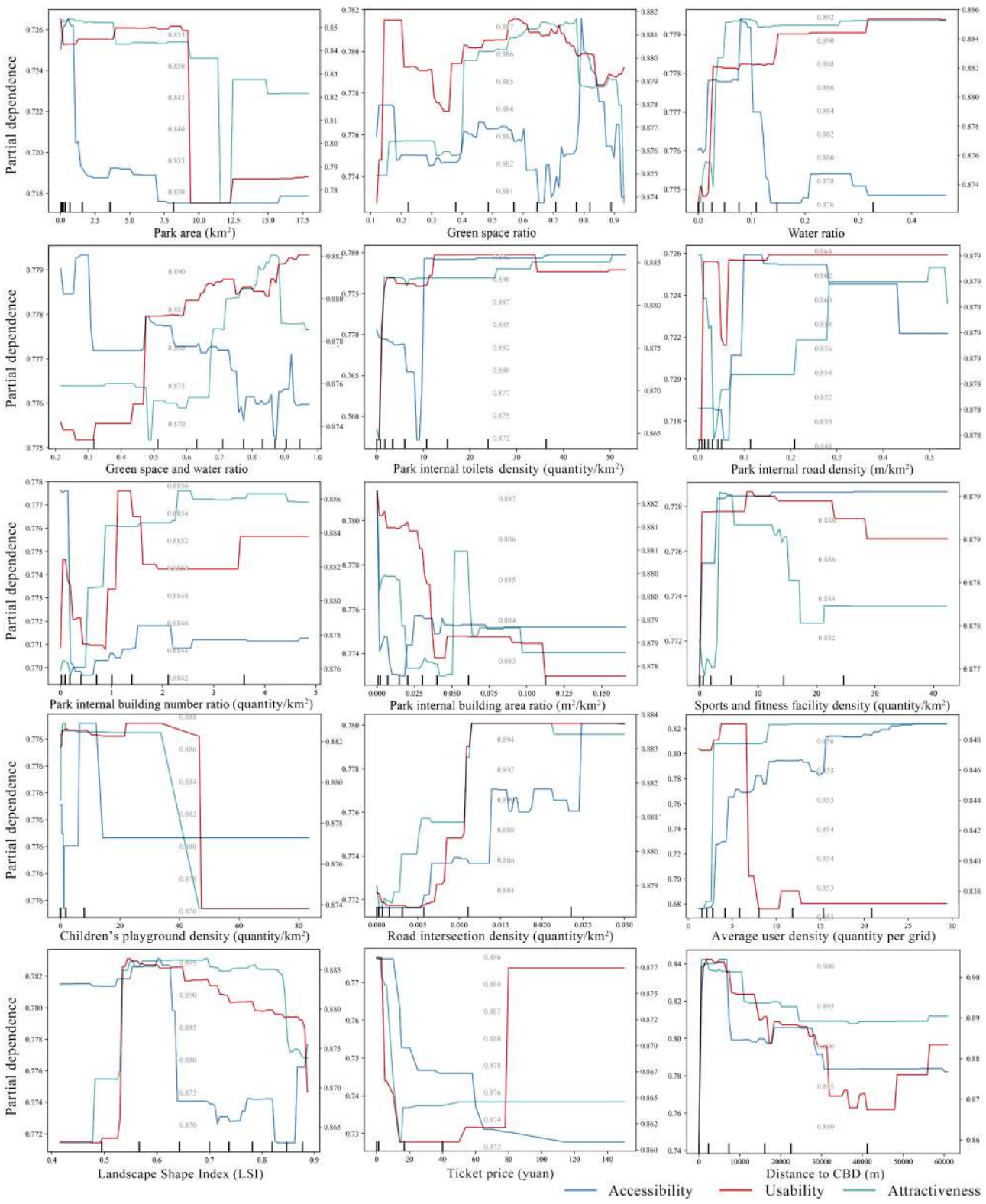


Fig. 8. Nonlinear influence of park interior attributes on park perceptions. Each plot features three y-axes, corresponding respectively to the partial dependence values for accessibility, usability, and attractiveness.

importance of easily navigable surroundings. The enclosure value does not markedly influence usability but is beneficial for accessibility and attractiveness. Conversely, high GVI and SVF values, often found in suburban areas, follow trends similar to those observed with remote sensing vegetation coverage and are generally less favored.

4. Discussion

4.1. The temporal and spatial patterns of park perceptions in Beijing and Guangzhou

Individuals in environments with positive perceptions are more

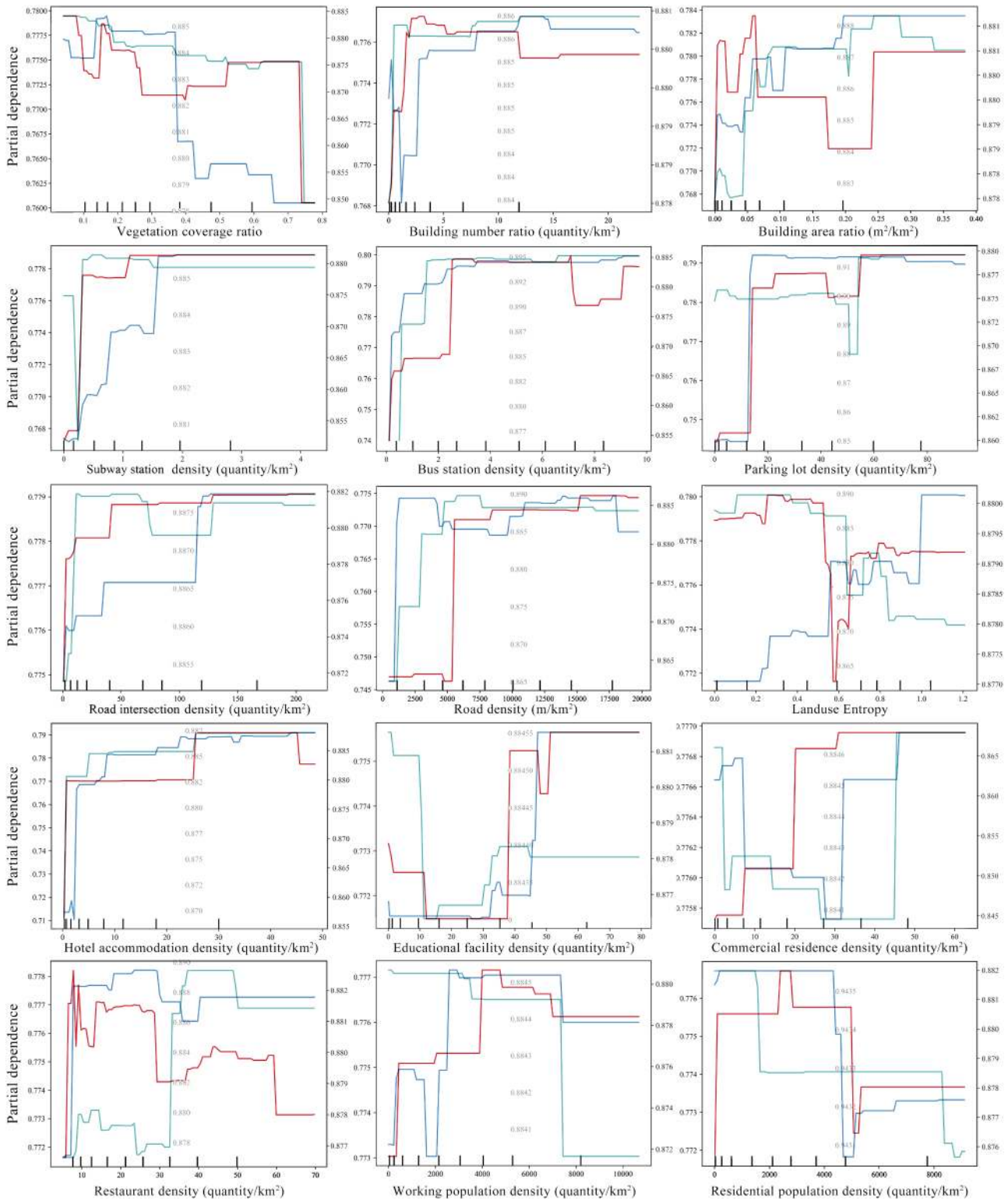


Fig. 9. Nonlinear influence of park surrounding attributes on park perceptions. Each plot features three y-axes, corresponding respectively to the partial dependence values for accessibility, usability, and attractiveness.

likely to engage in outdoor activities and adopt healthier lifestyles (Xu, Xiong et al., 2023), highlighting the need to assess temporal trends and spatial equality of park services from user-centered perspectives (Stessens, Canters et al., 2020).

Temporal analysis shows park visitation has a strong seasonal tendency, with cold winters in Beijing and hot summers in Guangzhou having relatively fewer visitors. Specially, both cities exhibit a significant increase in park visitation in October, coinciding with National Day's Golden Week in China, a period characterized by extended holidays that encourage outdoor leisure activities. However, this increase often results in overcrowded conditions in suburban parks, leading to a drop in perception scores (Tang, Zhao et al., 2023; Zhan, Guo et al.,

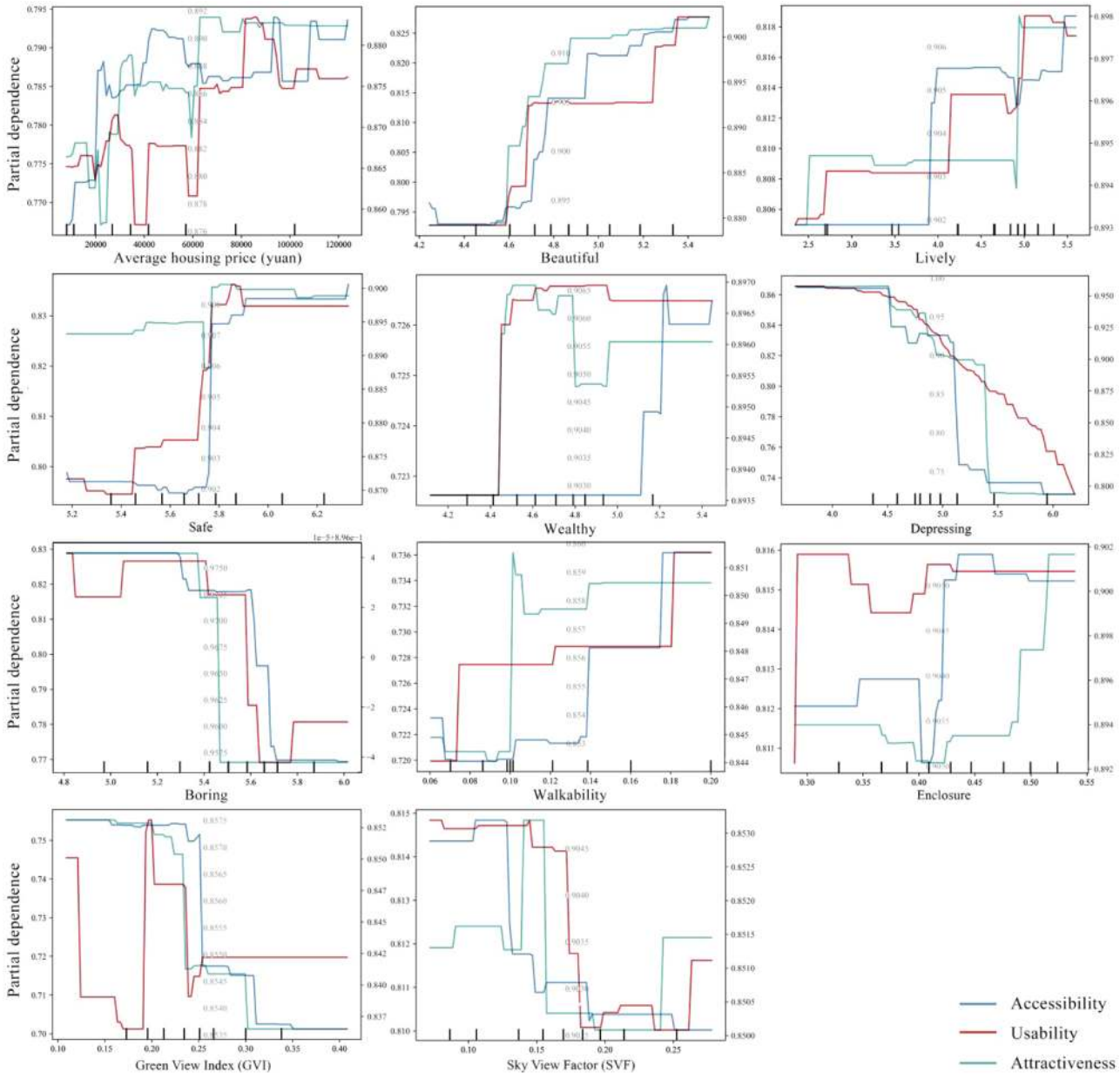


Fig. 9. (*continued*).

2024).

Both the monthly trend graphs and spatial distribution maps of the perception scores show that parks in central urban areas consistently outperform those in suburban regions. These parks maintain stable scores throughout the year, suggesting a resilient infrastructure capable of accommodating fluctuating visitor numbers and weather changes without a decline in park quality. This pattern aligns with research revealing that high-income populations in city centers often have better accessibility to high-quality parks (Biernacka, Łaszkiewicz et al., 2022; Zhang, Peng et al., 2022).

However, our application of the 2SFCA method incorporating perception scores reveals nuanced insights. Analysis under walking and riding modes indicates that subdistricts with high service capacities are primarily located in central urban areas, whereas the driving mode reveals expanded service to peripheral subdistricts, some of which display the highest service capacities. This distribution pattern likely results from the 2SFCA method's simultaneous consideration of population demand and park provision. Consequently, several suburban subdistricts with limited park access still exhibit high park service capacities due to lower population density. This indicates that residents near urban centers can be surrounded by more parks (Luo & Wang, 2003), but these parks may struggle to provide adequate services due to high population density.

Besides, the distribution of park services in central areas varies: Beijing shows a decline in service from west to east, whereas Guangzhou maintains high standards throughout its central areas. Although Beijing has a higher number of suburban parks with excellent scores, it also has more parks rated below 0.6 points, indicating areas that require significant attention. Enhancing the quality of existing parks, constructing new high-quality parks, or redeveloping parks from underutilized sites and decrepit infrastructure in areas where they are lacking could improve service equality (Li, Huang et al., 2021; Biernacka, Łaszkiewicz et al., 2022).

4.2. Effects of environmental characteristics on park perceptions

4.2.1. Effects of top-ranked attributes

Our study reveals that ticket prices significantly affect perceived accessibility, with higher prices correlating with poorer perception. Besides, previous research suggests that people's willingness to visit parks often hinges on the costs of time and distance, as well as the availability of transportation facilities (Wendel, Zarger et al., 2012; Wang, Brown et al., 2015). Our findings indicate that proximity to transportation stations, especially bus stations (Zhan, Guo et al., 2024), substantially enhances park accessibility. Other transportation-related factors, including distance to CBD, parking availability, road density, and subway station proximity, also play critical roles in improving access. Given the already well-developed public transport systems in Beijing and Guangzhou, shifting priorities towards enhancing subway facilities (Chen, Liu et al., 2018; Rui, 2023) and optimizing the road network (Li, Huang et al., 2021), especially in suburban areas, could further increase park accessibility.

For usability, research suggested that a balanced provision of well-designed outdoor amenities is important for physical activity and social interaction, which are primary motivations for visiting parks (Giles-Corti, Broomhall et al., 2005; Stessens, Canters et al., 2020; S. Huai, Liu et al., 2023). Similarly, our study indicates that sports and fitness facilities within parks have the highest influence on usability, with an optimal density identified as <22 per km^2. Additionally, the proximity of commercial residential areas such as apartments, communities, and villas were found to enhance usability, consistent with the prior studies indicating that many people enjoy walking in their local neighbourhood (Holman, Donovan et al., 1996). While average user density positively affects accessibility and attractiveness, its impact on usability is notably negative. Overcrowding and poorly maintained environments can create negative perceptions, potentially discouraging park use or even leading to avoidance (Wendel, Zarger et al., 2012; Zhan, Guo et al., 2024).

For attractiveness, proximity to CBD and the availability of bus stations are key factors. Being close to the CBD usually means better access to dense services, amenities, and transportation options, likely increasing incidental visits as visitors can engage in nearby recreational and catering services, thereby enriching their overall travel experience (Fan, Duan et al., 2021). Essential amenities within parks, such as toilets, the quantity of surrounding buildings, and restaurants also significantly enhance park experiences (Wendel, Zarger et al., 2012; Chen, Liu et al., 2018; S. Huai, Liu et al., 2023). Moreover, larger parks offer diverse environments and opportunities for visitors to "lose themselves" (Holman, Donovan et al., 1996) and generally have more attributes that make them attractive (Holman, Donovan et al., 1996; Giles-Corti, Broomhall et al., 2005). However, our results indicate that parks larger than 9 km^2 may experience significant declines in usability and attractiveness, necessitating targeted management strategies for large parks.

4.2.2. Additional novel insights

Apart from these factors with the highest impact, our study uncovers several novel insights. Classical literature has emphasized the importance of green and blue spaces in parks (Kaplan & Kaplan, 1989), and while these elements do not top our impact charts, they positively influence user perceptions to a certain extent. Interestingly, we identified a threshold effect around an 80 % green space ratio as the maximum value; exceeding this may have a detrimental impact on perceptions, suggesting that excessive greenery may not be favorable. This finding emphasizes the need for moderate and well-maintained greenery to enhance overall pleasantness and comfort (Xu, Xiong et al., 2023) and to decrease fear and insecurity (Liu, Li et al., 2023). Similarly, though facilities for children and park buildings can provide necessary social and psychological services (Chiesura, 2004; Wang, Brown et al., 2015), our results suggest that their presence should be limited.

Moreover, improving positive perceptions such as beauty, liveliness, safety, and wealth, as well as the walkability and enclosure of street environments around parks, is recommended (X. Zhao, Lu et al., 2024). These attributes are vital for creating a favorable first impression of a park and can significantly improve the overall park experience.

4.3. Research contributions

SMD reflects visitors' voluntarily shared opinions, providing a more genuine reflection of their thoughts without the potential biases introduced by close-ended survey questions (Wan, Shen et al., 2021). The Chinese-RoBERTa-wwm-ext model, fine-tuned on our manually curated dataset, has demonstrated robust capabilities in evaluating perceived park accessibility, usability, and attractiveness from massive SMD. Compared to traditional surveys and interviews, this approach enables faster and more accurate assessments of public perceptions across hundreds of parks, covering broader temporal and spatial scales. Unlike SMD research based solely on sentiment analysis, this approach facilitates precise evaluations of specific perception dimensions, offering finer-grained quantitative assessments of public attitudes.

Furthermore, integrating perception scores into the 2SFCA method balances park service supply with population demand while accounting for travel costs between parks and users. This comprehensive strategy helps identify underperforming areas that need enhancements in park quality or quantity. We also employed XGBoost and explainable machine learning algorithms to analyze the importance and nonlinear effects of various environmental attributes on park perceptions. These methods outperform linear analyses by better detecting dynamic trends and the impact magnitudes of influencing factors. Such user-centered insights are crucial for informing park design, planning, and policy interventions, contributing to the development of more livable and sustainable cities.

4.4. Research limitations and future directions

Firstly, our dataset focuses on popular parks that have accumulated a substantial number of comments on social media platforms. This scope excludes other green spaces such as grasslands, private gardens, neighborhood green spaces, or allotment gardens (Biernacka, Łaszkiewicz et al., 2022). Additionally, SMD might not fully represent all demographic groups, particularly the elderly and children. To mitigate these, supplementary surveys or alternative data sources could be used to capture broader perspectives.

Secondly, the Chinese-RoBERTa-wwm-ext model used in this study is an adaptation of Facebook AI's RoBERTa model (Liu, 2019) for Chinese NLP tasks. For other languages, the Facebook AI community provides alternative versions of RoBERTa (https://huggingface.co/FacebookAI). Researchers seeking to replicate our methods in other research disciplines or linguistic contexts can create targeted SMD datasets with relevant labels and fine-tune the appropriate RoBERTa model to assess various aspects of urban built environments. Additionally, the findings of this research may not be directly applicable to other countries or regions. Future studies could adopt our methodology and apply it to more diverse case study areas globally to develop more generalized insights.

Thirdly, while we considered a wide array of environmental attributes, not all potential influencing factors were included. Dynamic factors such as weather conditions and local events, which are more variable and difficult to assess, require further investigation. Besides, some factors that are challenging to assess through open-source data, such as the presence of birds (Holman, Donovan et al., 1996), perceived quietness (Stessens, Canters et al., 2020), and the historical and cultural significance of parks (Byrne & Wolch, 2009), were not included in our study. However, SMD still holds the potential to evaluate these aspects. Lastly, beyond environmental variables, examining the relationship between park perception and other variables such as socioeconomic variables (Giles-Corti, Broomhall et al., 2005) is both interesting and

crucial. Future research in these areas could provide valuable insights into the complex interactions between people and urban environments.

5. Conclusions

This study explores the temporal and spatial patterns and nonlinear driving mechanisms of park perceptions using advanced artificial intelligence algorithms and the 2SFCA method. We fine-tuned the Chinese-RoBERTa-wwm-ext model on a specially curated dataset to transform unstructured online comments into structured perception scores across three dimensions: accessibility, usability, and attractiveness. These scores enabled us to conduct statistical analyses to uncover temporal patterns and incorporate park perception scores into the 2SFCA method for spatial distribution analysis. Our findings indicate that parks in central urban areas consistently outperform those in suburban regions. Although residents living near urban centers enjoy higher accessibility to parks, some areas may struggle to provide adequate services due to high population density. These areas are suggested to enhance the quality of existing parks, construct new high-quality parks, or redevelop parks from underutilized sites and decrepit infrastructure.

We next investigated the nonlinear and threshold effects of environmental physical characteristics on park perceptions using XGBoost, SHAP, and PDP. The results reveal that accessibility is significantly influenced by ticket prices and the availability of transportation, especially bus stations. Usability is optimal at a moderate density of sports and fitness facilities (22 per km^2) and proximity to residential areas. Overcrowding and poorly maintained environments can create negative perceptions, potentially discouraging park use or even leading to avoidance. Attractiveness benefits from closeness to the CBD and available amenities such as bus stations, toilets, and restaurants, with a critical park size threshold of 9 km^2. Additionally, we note a threshold effect at an 80 % green space ratio, beyond which the benefits diminish. Enhancing positive perceptions, along with walkability and the enclosure of street environments around parks, is crucial for creating favorable first impressions and can significantly improve the overall park experience.

Overall, our workflow facilitates precise evaluations of perceived accessibility, usability, and attractiveness, allowing for the identification of parks or areas needing improvement while also detecting dynamic trends and the magnitudes of influencing factors. These user-centered approaches provide planners with valuable insights into public perceptions of urban parks, thereby supporting strategic decision-making and effective urban management. Future research should integrate SMD with investigative data to encompass a broader range of parks and explore possibilities for assessing more diverse perceptions from SMD.

CRediT authorship contribution statement

Xukai Zhao: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **He Huang:** Writing – review & editing, Validation. **Guangsi Lin:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization. **Yuxing Lu:** Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2024.106083.

Data availability

Data will be made available on request.

References

Abdelhamid, M. M., & Elfakharany, M. M. (2020). Improving urban park usability in developing countries: Case study of Al-Shalalat Park in Alexandria. *Alexandria Engineering Journal, 59*(1), 311–321.

Biernacka, M., Łaszkiewicz, E., & Kronenberg, J. (2022). Park availability, accessibility, and attractiveness in relation to the least and most vulnerable inhabitants. *Urban Forestry & Urban Greening, 73*, Article 127585.

Byrne, J., & Wolch, J. (2009). Nature, race, and parks: Past research and future directions for geographic research. *Progress in human geography, 33*(6), 743–765.

Chen, T., & Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*.

Chen, Y., Liu, X., Gao, W., Wang, R. Y., Li, Y., & Tu, W. (2018). Emerging social media data on measuring urban park use. *Urban Forestry & Urban Greening, 31*, 130–141.

Cheng, Y., Zhang, J., Wei, W., & Zhao, B. (2021). Effects of urban parks on residents' expressed happiness before and during the COVID-19 pandemic. *Landscape and Urban Planning, 212*, Article 104118.

Chiesura, A. (2004). The role of urban parks for the sustainable city. *Landscape and Urban Planning, 68*(1), 129–138.

Cui, Y., Che, W., Liu, T., Qin, B., & Yang, Z. (2021). Pre-training with whole word masking for Chinese bert. *IEEE/ACM Transactions on Audio, Speech, Language Processing, 29*, 3504–3514.

Dubey, A., Naik, N., Parikh, D., Raskar, R., & Hidalgo, C. A. (2016). Deep learning the city: Quantifying urban perception at a global scale. In *, Part I 14. Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings*. Springer.

Fan, Z., Duan, J., Lu, Y., Zou, W., & Lan, W. (2021). A geographical detector study on factors influencing urban park use in Nanjing, China. *Urban Forestry & Urban Greening, 59*, Article 126996.

Giles-Corti, B., Broomhall, M. H., Knuiman, M., Collins, C., Douglas, K., Ng, K., & Donovan, R. J. (2005). Increasing walking: How important is distance to, attractiveness, and size of public open space? *American Journal of Preventive Medicine, 28*(2), 169–176. ….

He, H., Sun, R., Li, J., & Li, W. (2023). Urban landscape and climate affect residents' sentiments based on big data. *Applied Geography, 152*, Article 102902.

Holman, C., Donovan, R. J., & Corti, B. (1996). Factors influencing the use of physical activity facilities: Results from qualitative research. *Health Promotion Journal of Australia: Official Journal of Australian Association of Health Promotion Professionals, 6*(1), 16–21.

Huai, S., Liu, S., Zheng, T., & Van de Voorde, T. (2023). Are social media data and survey data consistent in measuring park visitation, park satisfaction, and their influencing factors? A case study in Shanghai. *Urban Forestry & Urban Greening, 81*, Article 127869.

Huai, S., & Van de Voorde, T. (2022). Which environmental features contribute to positive and negative perceptions of urban parks? A cross-cultural comparison using online reviews and Natural Language Processing methods. *Landscape and Urban Planning, 218*, Article 104307.

Iwarsson, S., & Ståhl, A. (2003). Accessibility, usability and universal design—Positioning and definition of concepts describing person-environment relationships. *Disability and Rehabilitation, 25*(2), 57–66.

Kaplan, R., & Kaplan, S. (1989). *The experience of nature: A psychological perspective*. Cambridge university press.

Kong, I., Sarmiento, F. O., & Mu, L. (2023). Crowdsourced text analysis to characterize the US National Parks based on cultural ecosystem services. *Landscape and Urban Planning, 233*, Article 104692.

Kong, L., Liu, Z., Pan, X., Wang, Y., Guo, X., & Wu, J. (2022). How do different types and landscape attributes of urban parks affect visitors' positive emotions? *Landscape and Urban Planning, 226*, Article 104482.

Li, F., Li, F., Li, S., & Long, Y. (2020). Deciphering the recreational use of urban parks: Experiments using multi-source big data for all Chinese cities. *Science of the Total Environment, 701*, Article 134896.

Li, J., Fu, J., Gao, J., Zhou, R., Wang, K., & Zhou, K. (2023). Effects of the spatial patterns of urban parks on public satisfaction: Evidence from Shanghai, China. *Landscape Ecology, 38*(5), 1265–1277.

Li, J., Gao, J., Zhang, Z., Fu, J., Shao, G., Zhao, Z., & Yang, P. (2024). Insights into citizens' experiences of cultural ecosystem services in urban green spaces based on social media analytics. *Landscape and Urban Planning, 244*, Article 104999.

Li, X., Huang, Y., & Ma, X. (2021). Evaluation of the accessible urban public green space at the community-scale with the consideration of temporal accessibility and quality. *Ecological Indicators, 131*, Article 108231.

Li, Y., & Lin, G. (2024). Demand-side effects of urban green spaces: How attractiveness helps overcome subjective barriers to health behaviours. *Urban Forestry & Urban Greening, 94*, Article 128277.

Liu, Y. (2019). Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692.

Liu, Y., Li, Y., Yang, W., & Hu, J. (2023). Exploring nonlinear effects of built environment on jogging behavior using random forest. *Applied Geography, 156*, Article 102990.

Luo, W., & Wang, F. (2003). Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and planning B: Planning and Design, 30*(6), 865–884.

Ma, X., Ma, C., Wu, C., Xi, Y., Yang, R., Peng, N., & Ren, F. (2021). Measuring human perceptions of streetscapes to better inform urban renewal: A perspective of scene semantic parsing. *Cities (London, England), 110*, Article 103086. ….

Moran, M. R., Rodríguez, D. A., Cotinez-O'Ryan, A., & Miranda, J. J. (2020). Park use, perceived park proximity, and neighborhood characteristics: Evidence from 11 cities in Latin America. *Cities (London, England), 105*, Article 102817.

Nations, U. (2018). "2018 Revision of World Urbanization Prospects." Retrieved March 2024, from https://population.un.org/wup/.

Park, K. (2017). Psychological park accessibility: A systematic literature review of perceptual components affecting park use. *Landscape Research, 42*(5), 508–520.

Rui, J. (2023). Exploring the association between the settlement environment and residents' positive sentiments in urban villages and formal settlements in Shenzhen. *Sustainable Cities and Society, 98*, Article 104851.

Stessens, P., Canters, F., Huysmans, M., & Khan, A. Z. (2020). Urban green space qualities: An integrated approach towards GIS-based assessment reflecting user perception. *Land Use Policy, 91*, Article 104319.

Tang, Z., Zhao, Y., Fu, M., Wang, Y., & Xue, J. (2023). Which factors influence public perceptions of urban attractions?—A comparative study. *Ecological Indicators, 154*, Article 110541.

Tian, D., Wang, J., Xia, C., Zhang, J., Zhou, J., Tian, Z., & Zhou, C. (2024). The relationship between green space accessibility by multiple travel modes and housing prices: A case study of Beijing. *Cities (London, England), 145*, Article 104694. ….

Wan, C., Shen, G. Q., & Choi, S. (2021). Eliciting users' preferences and values in urban parks: Evidence from analyzing social media data from Hong Kong. *Urban Forestry & Urban Greening, 62*, Article 127172.

Wang, D., Brown, G., Zhong, G., Liu, Y., & Mateo-Babiano, I. (2015). Factors influencing perceived access to urban parks: A comparative study of Brisbane (Australia) and Zhongshan (China). *Habitat international, 50*, 335–346.

Wang, Z., Miao, Y., Xu, M., Zhu, Z., Qureshi, S., & Chang, Q. (2021). Revealing the differences of urban parks' services to human wellbeing based upon social media data. *Urban Forestry & Urban Greening, 63*, Article 127233.

Wang, Z., Zhu, Z., Xu, M., & Qureshi, S. (2021). Fine-grained assessment of greenspace satisfaction at regional scale using content analysis of social media and machine learning. *Science of the Total Environment, 776*, Article 145908.

Wei, D., Wang, Y., Liu, M., & Lu, Y. (2024a). User-generated content may increase urban park use: Evidence from multisource social media data. *Environment and Planning B: Urban Analytics and City Science, 51*(4), 971–986.

Wei, D., Wang, Y., Liu, M., & Lu, Y. (2024b). User-generated content may increase urban park use: Evidence from multisource social media data. *Environment and Planning B: Urban Analytics and City Science, 51*(4), 971–986.

Wendel, H. E. W., Zarger, R. K., & Mihelcic, J. R. (2012). Accessibility and usability: Green space preferences, perceptions, and barriers in a rapidly urbanizing city in Latin America. *Landscape and Urban Planning, 107*(3), 272–282.

Wenzhu, L., Enjia, Z., & Ying, L. (2024). Unveiling fine-scale urban third places for remote work using mobile phone big data. *Sustainable Cities and Society*, Article 105258.

Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J. M., & Luo, P. (2021). SegFormer: Simple and efficient design for semantic segmentation with transformers. *Advances in Neural Information Processing Systems, 34*, 12077–12090.

Xing, L., Liu, Y., Wang, B., Wang, Y., & Liu, H. (2020). An environmental justice study on spatial access to parks for youth by using an improved 2SFCA method in Wuhan, China. *Cities (London, England), 96*, Article 102405.

Xu, J., Xiong, Q., Jing, Y., Xing, L., An, R., Tong, Z., & Liu, Y. (2023). Understanding the nonlinear effects of the street canyon characteristics on human perceptions with street view images. *Ecological Indicators, 154*, Article 110756. ….

Yang, W., Li, Y., Liu, Y., Fan, P., & Yue, W. (2024). Environmental factors for outdoor jogging in Beijing: Insights from using explainable spatial machine learning and massive trajectory data. *Landscape and Urban Planning, 243*, Article 104969.

Zhan, P., Guo, Q., Chen, H., & Wu, Y. (2024). Antecedents and consequences of park crowding: Linking park attractiveness, perceived crowding, and revisit intention. *Landscape and Urban Planning, 245*, Article 105015.

Zhang, D., Ma, S., Fan, J., Xie, D., Jiang, H., & Wang, G. (2023). Assessing spatial equity in urban park accessibility: An improve two-step catchment area method from the perspective of 15-mintue city concept. *Sustainable Cities and Society, 98*, Article 104824.

Zhang, R., Peng, S., Sun, F., Deng, L., & Che, Y. (2022). Assessing the social equity of urban parks: An improved index integrating multiple quality dimensions and service accessibility. *Cities (London, England), 129*, Article 103839.

Zhao, X., Lu, Y., Huang, W., & Lin, G. (2024). Assessing and interpreting perceived park accessibility, usability and attractiveness through texts and images from social media. *Sustainable Cities and Society, 112*, Article 105619.

Zhao, X., Lu, Y., & Lin, G. (2024). An integrated deep learning approach for assessing the visual qualities of built environments utilizing street view images. *Engineering Applications of Artificial Intelligence, 130*, Article 107805.

Zheng, T., Pan, Q., Zhang, X., Wang, C., Yan, Y., & Van De Voorde, T. (2024). Research Note: Linking sensory perceptions with landscape elements through a combined approach based on prior knowledge and machine learning. *Landscape and Urban Planning, 242*, Article 104928.

Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., & He, Q. (2020). A comprehensive survey on transfer learning. *Proceedings of the IEEE, 109*(1), 43–76. ….

Zube, E. H., Sell, J. L., & Taylor, J. G. (1982). Landscape perception: Research, application and theory. *Landscape Planning, 9*(1), 1–33.